CampusNotes - Academic Study Resource

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING - SEMESTER 5

OFFICIAL ACADEMIC STUDY MATERIAL - CSE CURRICULUM

Experiment 3: Data Aggregation with Java 8 Streams & Lambdas

Course: Advanced Java Programming Laboratory [23CS3504]
Category: Experiments / Practicals | Experiment #3 | Academic Year: 2026-27

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1. Overview & Objective:

Process and filter large dataset collections using declarative Stream pipelines.

2. Practical Tasks & Verification Steps:

1. Model Student record with attributes: id, name, semester, branch, cgpa, and attendance.
2. Filter all CSE Sem 5 students with CGPA >= 8.5 and sort them alphabetically using lambda comparators.
3. Compute branch-wise average CGPA using Collectors.groupingBy and Collectors.averagingDouble .
4. Demonstrate parallelStream execution speedup on a synthetic dataset of 500,000 records.

Submission Guidelines & Verification:

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